

Interactive Visualization of Dynamic Functional Connectivity in the ABIDE Autism Dataset

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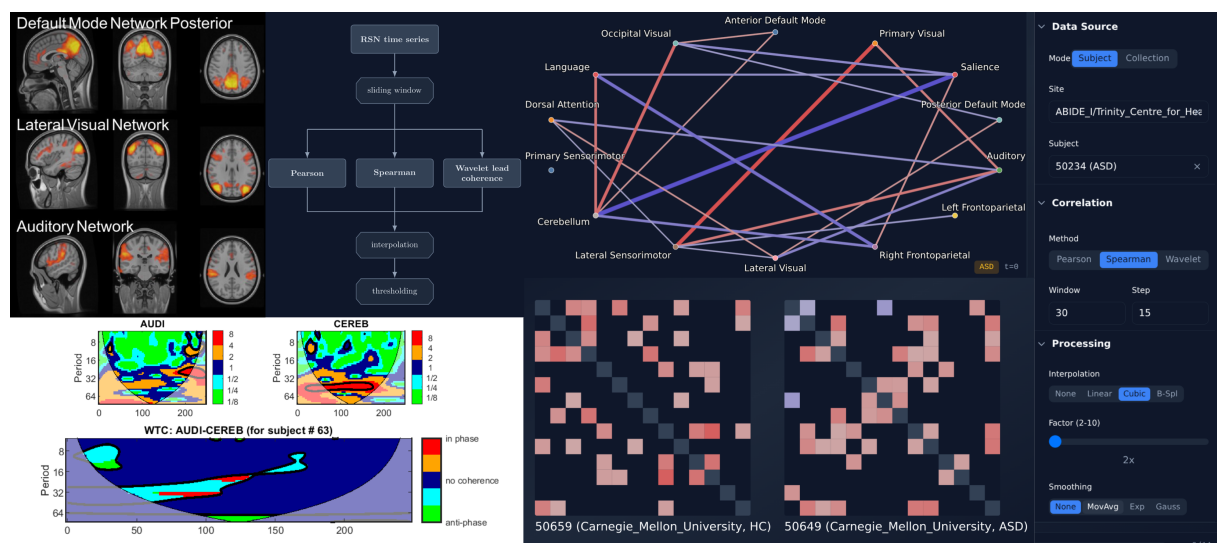


Figure 1: Windowed connectivity between the 14 resting state networks, computed per subject and explored through an interactive graph view and a comparative overview across diagnostic classes and sites.

Abstract

Autism spectrum disorder (ASD) is currently diagnosed using observation-based criteria, while an objective test is desired [SPA24]. Resting state functional connectivity between brain regions is increasingly understood to be a dynamic process, varying in time [HWA*13, PBVDV16]. However, the tools currently available collapse a recording into a single correlation matrix and discard the time variations. We present an interactive web application for exploring the time varying functional connectivity between brain regions in the large ABIDE I+II autism fMRI dataset. Each recording is decomposed into resting state networks whose dynamic connectivity is measured over a sliding window using several methods, including wavelet lead coherence, a directional measure of which network leads which. These connections are visualized as a video of a graph with weighted edges, alongside an overview pane to compare individual scans, or entire diagnostic classes. Our system turns a large, otherwise opaque dataset into a visual format for discovering time varying functional connectivity patterns. It is a step toward an interpretable neuroimaging tool that may aid diagnosis.

CCS Concepts

• Information systems → Information systems applications → Decision support systems → Data analytics

1. Introduction

Resting state fMRI (rs-fMRI) is the most studied imaging method in autism spectrum disorder (ASD) research [SPA24], and the functional connectivity it reveals is central to studying the condition. This connectivity is conventionally computed over an entire recording, resulting in a single static connectivity matrix. However, dynamic functional connectivity captures information that static correlations discard [CG09], and has shown promise as a biomarker for the diagnosis of autism [BAZ17], depression [CPB*23, PLS*24], and cognitive aging [BBAZ21].

The Autism Brain Imaging Data Exchange (ABIDE) is a large, multisite rs-fMRI dataset. Recent work preprocessed ABIDE I+II using independent component analysis (ICA) into resting state networks (RSNs) for 900 subjects (417 ASD, 483 controls) [SPA*25]. ABIDE is the most comprehensive fMRI ASD dataset [SPA24], which is ideal for this dynamic functional connectivity visualization. However, common connectivity visualizers are static and lack the interactive, time varying exploration needed to understand dynamic patterns. We present a novel web application that fills this gap with an animated graph view of windowed connectivity between 14 RSNs, computed per subject from configurable parameters. It computes this connectivity using symmetric correlations and a directional wavelet lead coherence measure, and provides a comparative overview across diagnostic classes and specific sites. Figure 1 shows an overview of the system, from the RSNs, through the windowed connectivity correlations, to the interactive graph and overview views.

2. Neuroimaging and Brain Connectivity

From group independent component analysis (ICA), 14 resting state networks were identified [SPA*25]. Functional connectivity between two networks is quantified by correlating their time series. Applied over a fixed length sliding window, this produces a series of matrices that describe the dynamic functional connectivity [HWA*13, PBVDV16].

The system offers three correlation measures. Pearson and Spearman correlation produce symmetric matrices in the range $[-1, 1]$. The third is a directional (asymmetric) measure, called wavelet lead coherence. The wavelet transform decomposes each signal into time and frequency, capturing how its frequency content varies over time rather than averaging over the whole scan. Wavelet coherence then acts as a localized correlation coefficient between two signals, and its cross-wavelet phase angle gives their lead or lag at each point in time and frequency [GMJ04, TC98, CG09], classifying each network pair as in-phase, anti-phase, leading, or lagging [BAZ17, CPB*23]. The wavelet lead coherence measure reports values in the range $[0, 1]$, the fraction of time and frequency points at which one network leads the other, computed in both directions, so the resulting matrix

is asymmetric. This directional measure of correlation highlights lead and lag patterns that are consistent with causal influence and may motivate further research into causal relationships between networks [CPB*23].

3. Visualizations of Dynamic Brain Activity

The graph view places the resting state networks on a ring and draws the current frame's connectivity matrix as edges (Figure 1). The color of an edge encodes the correlation value on a blue to red scale and thickness encodes its absolute magnitude. A configurable threshold is used to hide weak edges and denoise the visualization. For wavelet lead coherence, the edges carry arrows showing the directionality of the correlation. The frames are animated over time, showing how the graph changes. Hovering over a network node isolates its edges, and hovering over an edge shows the two networks it connects and their correlation value. This view can also play aggregate sequences, the mean or median over all ASD or all control subjects, so an individual's dynamics can be compared against either class. Any change in the parameters (window size, step length, correlation measure, smoothing, interpolation, and thresholding) recomputes the graph in real time, enabling users to explore dynamic connectivity patterns quickly.

A complementary overview pane (Figure 1) reduces each subject to a single correlation matrix over the entire recording, shown as a heatmap or a miniature graph, separated into healthy control and ASD columns and grouped by acquisition site. Read against the class mean and median, the overview enables fast scanning for individual outliers, group contrasts, and site variation. Identifying site variation and measurement error induced outliers is especially important, as it is a known confound in multisite rs-fMRI [SPA24].

4. Conclusions

Exploring dynamic functional connectivity of RSNs is a promising method for discovering explainable diagnostic features for the objective diagnosis of conditions such as ASD, but the dynamic patterns are difficult to observe with the static visualization tools currently available. Our contribution is a novel visualization tool for observing dynamic functional connectivity patterns between RSNs, publicly available at <https://brainviz.ideabattery.org/>. Our system visualizes the large autism fMRI dataset ABIDE I+II, enabling interactive exploration of each subject's RSN connectivity over time. This is a step toward translating high-dimensional fMRI features into interpretable information that can aid clinicians and produce explainable features for training machine learning models. Planned work adds a quantitative comparison of ASD against control subjects, turning exploratory observations into testable group differences and diagnostic features.

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